Automotive Audio Bus A²B Transceiver

Silicon Anomaly List

AD2435/AD2433/AD2432/AD2431

ABOUT AD2435/AD2433/AD2432/AD2431 SILICON ANOMALIES

These anomalies represent the currently known differences between revisions of the A²B® AD2435/AD2433/AD2432/AD2431 product(s) and the functionality specified in the AD2435/AD2433/AD2432/AD2431 data sheet(s) and the Technical Reference manual.

SILICON REVISIONS

A silicon revision number with the form "-x.x" is branded on all parts. The silicon revision can be electronically determined by reading the **A2B_VERSION** and **A2B_PRODUCT** registers.

Silicon REVISION	A2B_VERSION[7:0]	A2B_PRODUCT[7:0]	PRODUCT
1.1	0x11	0x31, 0x32, 0x35	AD2431, AD2432, AD2435
1.0	0x10	0x33	AD2433

ANOMALY LIST REVISION HISTORY

The following revision history lists the anomaly list revisions and major changes for each anomaly list revision.

Date	Anomaly List Revision	Data Sheet Revision	Additions and Changes
09/29/2020	B	PrD	Added Silicon Revision 1.0 and 1.1 Added Anomalies: 18000045, 18000046, 18000047, 18000048, 18000049, 18000050
12/13/2019	A	PrC	Initial Version

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SUMMARY OF SILICON ANOMALIES

The following table provides a summary of AD2435/AD2433/AD2432/AD2431 anomalies and the applicable silicon revision(s) for each anomaly.

No.	ID	Description	Rev 1.0	Rev 1.1
1	18000045	Short to VBAT Fault May Be Incorrectly Reported When LVI Mode Is Enabled In CFG-0	x	.
2	18000046	Pseudo-Random Binary Sequence Test Mode May Report Erroneous Bit Errors	x	x
3	18000047	ADR1/CLKOUT1 Or ADR2/CLKOUT2 Is Not Functional For Certain GPIO Modes When SPI Is Enabled	x	x
4	18000048	BP Short To Ground Fault Incorrectly Reported Under Certain Conditions	x	.
5	18000049	BP short to VBAT Fault May Not Be Detected After Discovery in CFG-0	x	.
6	18000050	Partial Re-discovery Of A Slave Node Corrupts The Slots Of Upstream Nodes	x	x

Key: x = anomaly exists in revision
. = Not applicable

DETAILED LIST OF SILICON ANOMALIES

The following list details all known silicon anomalies for the AD2435/AD2433/AD2432/AD2431 including a description, workaround, and identification of applicable silicon revisions.

1. 18000045 - Short to VBAT Fault May Be Incorrectly Reported When LVI Mode Is Enabled In CFG-0:

DESCRIPTION:
When a master or Locally Powered Slave (LPS) operating in LVI mode is driving a Bus Powered Slave (BPS), then a short to VBAT on B-Port of the BPS may be incorrectly reported as short to GND at the master or LPS. This is applicable only when the master or LPS is operating in CFG-0.

WORKAROUND:
Use any one of the workarounds listed below.

1. Configure the Master or LPS node driving the BPS node in non-LVI mode.
2. If short to GND at Master or LPS node is detected, then host can perform the below actions:
   a. Write A2B_SWCTL.ENSW = 0 in the node where the fault occurred and wait for 100ms.
   b. Start discovery from the same node with A2B_SWCTL = 0x21 for every CFG-0 node.
   c. Continue if discovery is successful (A2B_INTTYPE = 0x18) or report the error.

APPLIES TO REVISION(S):
1.0

2. 18000046 - Pseudo-Random Binary Sequence Test Mode May Report Erroneous Bit Errors:

DESCRIPTION:
The Pseudo-Random Binary Sequence (PRBS) test mode may report erroneous bit errors when SPI Data Tunnel is enabled and if the data tunnel slots are present in Slot 0 and Slot1. The PRBS mode stops the remote SPI access through the data tunnels.

WORKAROUND:
If SPI data tunnel and PRBS mode are used in conjunction, the SPI data tunnel downstream offset (**A2B_DTDNOFFS . DTDNOFFS**) should be >=2

APPLIES TO REVISION(S):
1.0, 1.1

3. 18000047 - ADR1/CLKOUT1 Or ADR2/CLKOUT2 Is Not Functional For Certain GPIO Modes When SPI Is Enabled:

DESCRIPTION:
ADR1/CLKOUT1 and ADR2/CLKOUT2 can be used to source the clock to the peripherals. When SPI is enabled then any one of the ADR1/CLKOUT1 or ADR2/CLKOUT2 is not functional based on the GPIO mode.

1. ADR1/CLKOUT1 is not functional when SPI is enabled with Non-Default GPIO select(GPIO MODE=2/3).
2. ADR2/CLKOUT2 is not functional when SPI is enabled with Default GPIO select(GPIO MODE=0).

The table below explains ADR1/CLKOUT1 and ADR2/CLKOUT2 availability based on SPI configuration and GPIO modes.

{ADR1, ADR2} Pin Functioning as CLKOUT[1,2]

SPI Configuration		GPIO Selection			
SPI Mode	Chip Select	GPIO Mode 0 (on SPI Pins)		GPIO Mode 2/3 (on I²C and Rx/Tx Pins)	
		CLKOUT1	CLKOUT2	CLKOUT1	CLKOUT2
SPI Slave	ADR1/$\overline{\text{SPISS}}$	N/A	x	N/A	ADR2
	SIO2/$\overline{\text{ASPISS}}$	x	x	x	ADR2
	ADR2/$\overline{\text{ASPISS}}$	x	N/A	x	N/A
SPI Master	ADR1/$\overline{\text{SPISSEL0}}$	N/A	x	N/A	ADR2
	SIO2/$\overline{\text{SPISSEL1}}$	ADR1	x	x	ADR2
	ADR2/$\overline{\text{SPISSEL2}}$	ADR1	N/A	x	N/A

[1] N/A denotes not applicable.
[2] x denotes not functional.

WORKAROUND:
Use any one of the workarounds listed below.

1. Use non-default GPIO mode (A2B_PINCFG.GPIOSEL != 0) to source the clock to peripheral with ADR2/CLKOUT2 when SPI is enabled.
2. Use default GPIO mode (A2B_PINCFG.GPIOSEL = 0) to source the clock to peripheral with ADR1/CLKOUT1 when SPI is enabled.
3. Disable SPI to use both ADR1/CLKOUT1 and ADR2/CLKOUT2.

APPLIES TO REVISION(S):
1.0, 1.1

4. 18000048 - BP Short To Ground Fault Incorrectly Reported Under Certain Conditions:

DESCRIPTION:
When master or Locally Powered Slave (LPS) operating in non-LVI mode is driving an AD240x/AD241x/AD242x Bus Powered Slave (BPS), a BP short to ground fault on B-Port of the BPS may be incorrectly reported as wires shorted together at Master or LPS. This is applicable only when the master or LPS is operating in CFG-0.

WORKAROUND:
Use any one of the workarounds listed below.

1. Configure the AD243x upstream node driving the AD240x/AD241x/AD242x BPS node in LVI mode.
2. If BP short to BN at upstream node is detected, then host can perform the below actions:
   a. Write A2B_SWCTL.ENSW = 0 to the upstream node (Master or LPS) and wait for 100ms.
   b. Start discovery with A2B_SWCTL = 0x21.
   c. Continue if discovery is successful (A2B_INTTYPE = 0x18)or report the error.

APPLIES TO REVISION(S):
1.0

5. 18000049 - BP short to VBAT Fault May Not Be Detected After Discovery in CFG-0:

DESCRIPTION:
When the transceiver is operating in CFG-0, BP short to VBAT fault may not be detected after discovery. This is applicable for both LVI and non-LVI mode.

WORKAROUND:
Use the below mentioned flow.

1. Enable VMTR (Voltage Monitor) ADC to monitor VBUS voltage by writing A2B_VMTR_VEN = 0x02
2. Enable VMTR interrupt for VBUS channel by writing A2B_VMTR_INTEN = 0x02.
3. Set threshold for generating max interrupt by using A2B_VMTR_VMAX1 register.
   eg: Set A2B_VMTR_VMAX1 = 0x48 to generate interrupt whenever VBUS goes above 9V
4. Initiate a rediscovery when A2B_INTTYPE = 0x50 & A2B_VMTR_MXSTAT = 0x02 condition is satisfied

APPLIES TO REVISION(S):
1.0

6. 18000050 - Partial Re-discovery Of A Slave Node Corrupts The Slots Of Upstream Nodes:

DESCRIPTION:
If a slave node is disconnected from the A2B bus, host can attempt to rediscover the dropped node. Consider an A2B network as Master - Slave0 - Slave1, with Slave0 and Slave1 contributing to 2 upstream slots. If Slave1 is disconnected from the bus, then the communication can happen between Master and Slave0, with 2 slots of Slave1 replaced with previous known good samples.

When host initiates a partial rediscovery of Slave1, upon successful discovery it results in audio distortion due to corruption of the slots contributed by Slave0. This is because the Slave1 is just discovered and not yet configured. Once the Slave1 is successfully configured, no further distortion is observed.

WORKAROUND:
Use any one of the workarounds listed below.

1. Host can configure the A2B_LUPSLOTS, A2B_UPSLOTS, A2B_SLOTFMT, A2B_DATCTL, A2B_CONTROL.NEWSTRUCT registers initially so as to reduce the duration of noise. Other registers can be configured later
2. Before initiating the partial rediscovery, the A2B_UPSLOT register of last good upstream node (Slave0 in this example) can be programmed to 0 and map the TDM channels accordingly. Host can reconfigure to actual values upon successful rediscovery (Slave1 in this example)

APPLIES TO REVISION(S):
1.0, 1.1

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